Paleomagnetism of Permian Redbeds from the Colorado Plateau

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Partial thermal demagnetization of Permian sediments from two locations on the Colorado Plateau has, in some cases, isolated a thermally stable paleomagnetic direction that is significantly different from the results previously published for the Permian rocks of the southwestern United States. The stable magnetization was successfully recovered from samples of the Halgaito tongue of the Cutler formation, the Hoskinnini tongue of the Moenkopi (Triassic?) formation, the Toroweap formation, and the Hermit shale. The NRM of other formations, including the Supai, was found to be thermally unstable. Thermal demagnetization shows that a secondary component of magnetization parallel to the earth's present dipole field direction has probably been acquired by all the Permian sediments of this area.

INTRODUCTION

Several paleomagnetic studies have been published on the natural remanent magnetization (NRM) of Permian sediments from the southwest United States [*Graham*, 1955; *Runcorn*, 1955, 1956; *Doell*, 1955; *Collinson and Runcorn*, 1960], but there have been no systematic investigations of the magnetic stability of these rocks. In some cases it was known from a streaked distribution of the NRM that a secondary component of magnetization had been acquired in the earth's present magnetic field, and, in such instances where streaking of the NRM vectors was obvious, it was customary when calculating a virtual magnetic pole to include only the points farthest removed from the present field direction. This is at best an uncertain procedure to correct for the recent magnetization and has been shown [*Irving et al.*, 1961] to give erroneous results.

With the intention of isolating a stable primary magnetization by thermally removing the secondary components, three collections were made from flat-lying redbeds on the Colorado Plateau:

1. 120 samples from 7 sites in the Cutler formation and the beds immediately above and below it in the Monument Valley area.
2. 148 samples from 2 sites in Grand Canyon National Park.
3. 37 samples from 3 sites in northwestern Arizona.

Figure 1 shows the geographic location of the two principal collecting areas and Figure 2 their approximate stratigraphic correlation. See Table 1 for details of site locations and lithology. From these samples some 450 specimens were measured and most specimens were heated to test for thermal stability.

EXPERIMENTAL METHODS

The magnetic stability of rocks is best studied through a combination of laboratory and field tests (see, for instance, *Irving* [1964, pp. 73–93] for discussion on paleomagnetic reliability tests), but the field tests are generally not applicable to the sediments in this area, although *Graham* [1955] has reported an inconclusive application of the conglomerate test. Because the magnetic particles in redbeds usually have a high coercive force (>1000 gauss), the most suitable laboratory method for investigating the stability of the NRM is progressive thermal demagnetization. In this technique, the paleomagnetic samples are heated to successively higher temperatures, cooled in a field free space (<200 γ) and remeasured. Since the magnetic moment of a given grain has a relaxation time that depends exponentially on temperature [*Néel*, 1955], the

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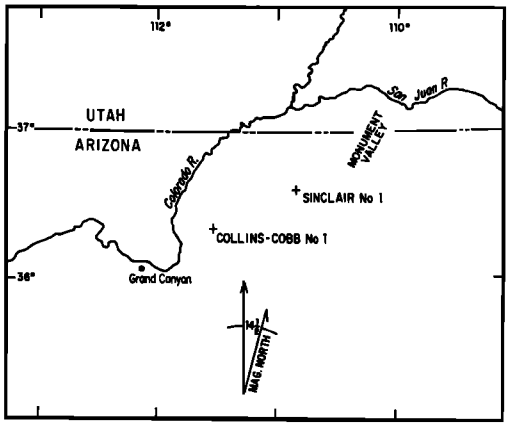


Fig. 1. Location map showing Grand Canyon and Monument Valley areas and two wells used in the stratigraphic correlation.

successive heatings isolate the magnetized grains with longer and longer relaxation times. In this way, the NRM vector is analyzed into constituent vectors, each with a different time constant.

Chamalaun [1963] studied the thermal demagnetization of a wide variety of red sediments. He found that the furnace should be open to the atmosphere, because heatings made in a nitrogen environment caused reduction of the red cement. A similar conclusion was reached by *Leng* [1955] and *Irving and Opdyke* [1965]. Even when the heatings are done in air, the high temperatures produce physicochemical changes in the magnetic minerals, for there is a marked increase in the room temperature susceptibility and the rocks often acquire an IRM if left in an ambient magnetic field [*Chamalaun*, 1963; *Chamalaun and Creer*, 1964]. These changes do not, however, seem to affect the most stable magnetization, and Chamalaun and Creer applied this technique to the Devonian Old Red Sandstone, removing a Permo-Carboniferous component of magnetization. *Irving and Opdyke* [1965] have used the method to remove a Permian component of magnetization from the Silurian Bloomsburg redbeds, and *Briden* [1965] has reported work

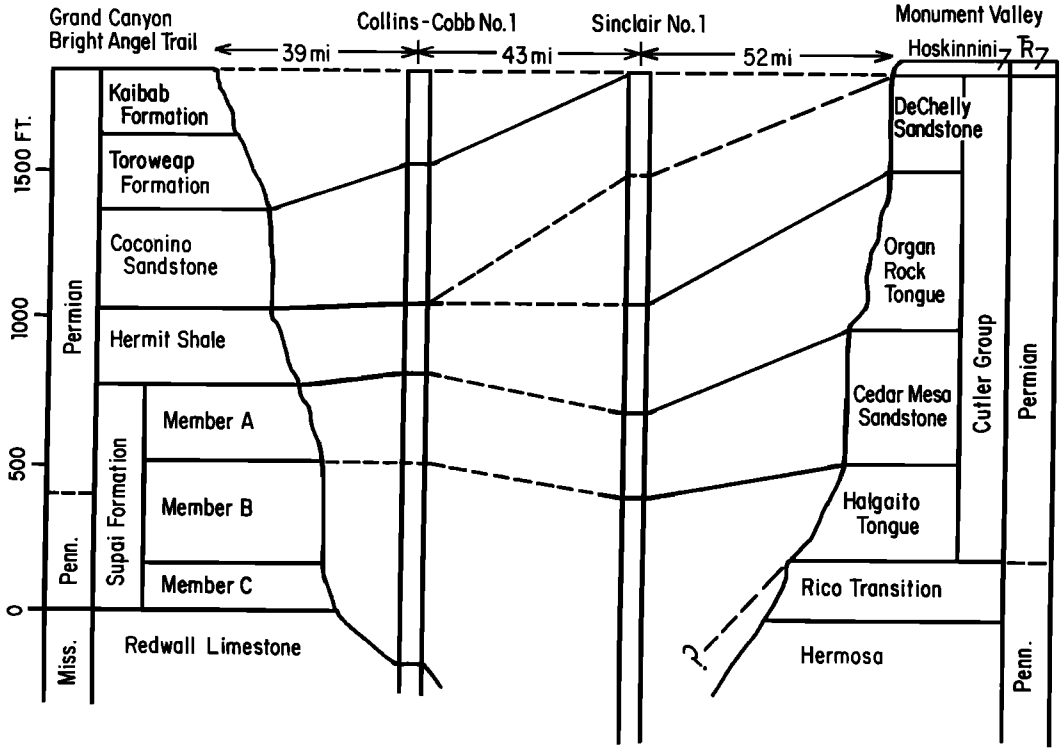


Fig. 2. Stratigraphic correlation between Grand Canyon and Monument Valley sections (H. W. Peirce, personal communication).

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TABLE 1. Sampling Data

Formation	Sampling Locations	No. of Samples	Lithology
Hoskinnini tongue (Triassic?)	36.9°N, 110.3°W	20	Brown, irregularly bedded gritty sandstone.
DeChelly sandstone	36.9°N, 110.3°W	20	Brownish-orange, fine-medium grained, cross-bedded. Massive quartzose sandstone.
Organ Rock tongue	37.0°N, 110.1°W	20	Alternating brownish-orange sandstone, siltstone, and limestone.
Cedar Mesa sandstone	37.3°N, 109.9°W	20	Gray or brown alternating sandstone, siltstone, and limestone.
Halgaito tongue	37.2°N, 109.8°W	20	Brownish-orange siltstone, sandstone, and limestone.
Rico formation	37.1°N, 109.9°W	20	Marine, alternating calcareous sandstone, gray and red shale, and limestone.
Toroweap formation	36.1°N[a], 112.1°W	11	Red and yellowish sandstones with intermediate marine gray limestone.
	36.1°N[b], 112.3°W	11	
Hermit shale	36.1°N[a], 112.1°W	17	Reddish-brown shales and siltstones.
	36.1°N[b], 112.3°W	8	
Supai formation	36.1°N[a], 112.1°W	75	Thick series of alternating red cross-bedded sandstones and shales.
	36.1°N[b], 112.3°W	20	
	35.4°N[c], 112.9°W	12	
	35.7°N[d], 113.5°W	13	
	36.3°N[e], 113.8°W	12	

[a] Kaibab Trail, Grand Canyon National Park.
[b] Hermit Trail, Grand Canyon National Park.
[c] Aubrey Cliffs.
[d] Quartermaster Viewpoint.
[e] Grand Wash Cliffs.

done on a variety of Paleozoic rocks from Australia.

The measurements of the Hermit Trail samples from the Toroweap formation and Hermit shale were made on a spinner magnetometer [*DeSa and Molyneux*, 1963]; for all other samples an astatic instrument [*Collinson et al.*, 1957] was used. The furnace and its ancillary apparatus were those described by *Chamalaun* [1963]. Samples were either oriented hand samples or oriented cores taken with a portable drill. The uncertainty in any one measurement, including collecting errors, is estimated to be ±4° in direction and ±10% in intensity.

Monument Valley Results

NRM directions for the Monument Valley formations (Figure 3) show two distribution types, one centered near the present axial dipole direction and the other scattered or 'streaked' from the present field direction toward the southeast. These results, similar to the results found in previous studies in this area, indicate that the NRM consists of a primary magnetization with a superimposed secondary component and that the ratio of secondary to primary intensities varies greatly from specimen to specimen and formation to formation.

Three pilot specimens representing various NRM directions and intensities were picked from each formation and progressively demagnetized. Two examples of the demagnetization curves, plots of direction and intensity versus temperature, are presented in Figure 4. The remanence in specimen 2 exhibits a large decrease in intensity and erratic change in direction with each heating. The plateau in the intensity curve at 600°C hints at the presence of two magnetizations in the specimen, but the continual change in direction with each heating makes it impossible to isolate a unique thermally stable direction. If such rocks contain any ancient magnetization, it is not stable and cannot be separated from the more recent magnetization. These specimens are useless for determining paleomagnetic field directions.

The demagnetization curves for specimen 1 show, however, that the remanence consists of a

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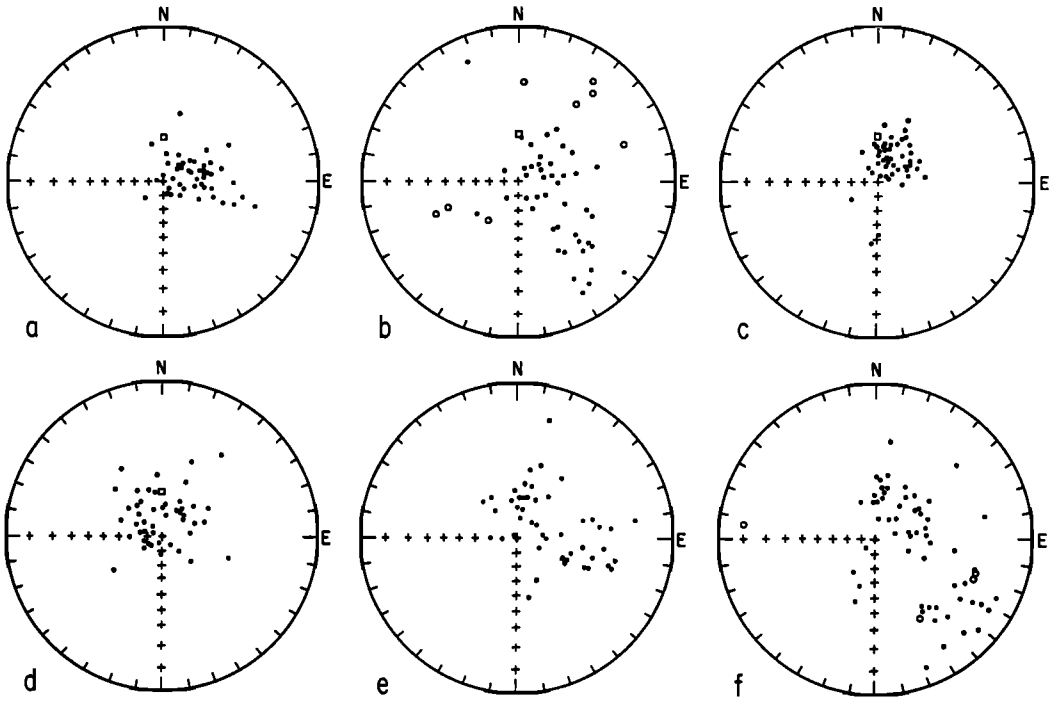


Fig. 3. NRM observed in Monument Valley rocks: (*a*) Rico transition, (*b*) Halgaito tongue, (*c*) Cedar Mesa sandstone, (*d*) Organ Rock tongue, (*e*) DeChelly sandstone, (*f*) Hoskinnini tongue. Squares indicate geocentric dipole; open circles, upper hemisphere; solid circles, lower hemisphere. These symbols are the same for all figures.

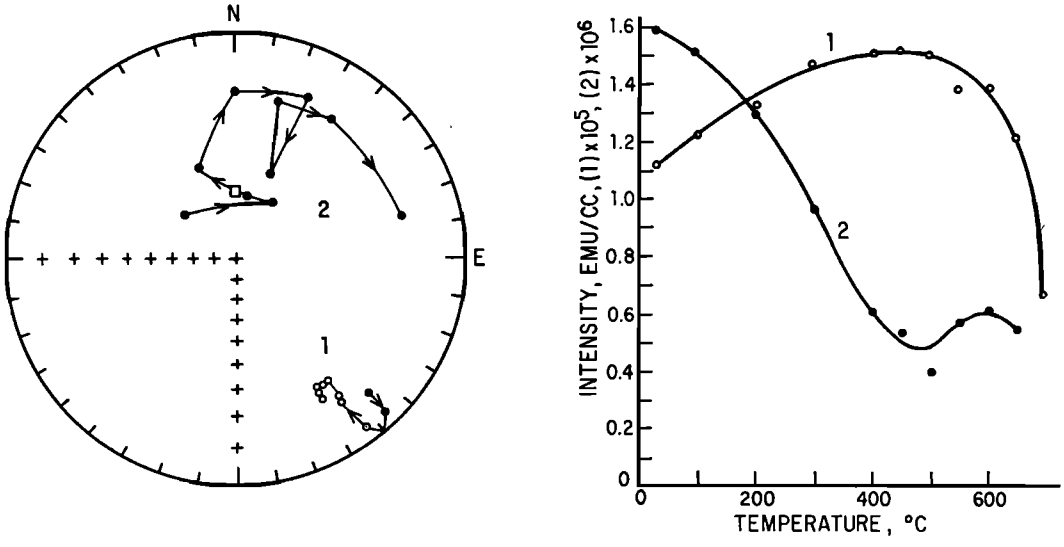


Fig. 4. Two thermal demagnetization curves exhibiting extreme examples of stable (specimen 1) and unstable (specimen 2) magnetizations. Note the different intensity scales.

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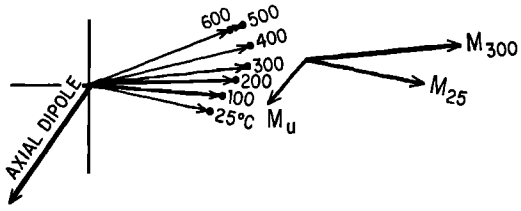


Fig. 5. The residual magnetization vectors for specimen 1 projected onto the plane of streaking. M_u is the component of magnetization destroyed by heating to 300°C. A heating to 500°C was required to remove completely all secondary magnetization.

thermally stable and a thermally unstable component, the thermally unstable component being destroyed by temperatures of ≃500°C. The direction, which changes regularly over the first few heatings as the unstable magnetization is eliminated, remains essentially constant above 500°C. The reproducibility of the high-temperature magnetization over three heatings between 500° and 600°C indicates that the unstable component has been destroyed.

The thermally stable magnetization in specimen 1, the NRM, and the axial dipole direction are nearly coplaner and define the plane of streaking. Figure 5, a plot versus temperature of the vector magnetization projected onto the plane of streaking, shows that up to 300°C the magnetization removed (M_u) was accurately parallel to the axial dipole direction. It is probable that the components destroyed at 400° and 500°C were also parallel to the axial dipole direction, but changes in the intensity of the stable magnetization at these temperatures obscured the true relation. An approximate parallelism between the present field direction and the unstable magnetization was observed in numerous specimens over the first few heatings.

The relative intensity at high temperatures of all pilot specimens that possessed a thermally stable direction was significantly greater than the relative intensity of the specimens whose direction changed with each heating. If a pilot specimen contained a thermally stable component, it was isolated from the unstable components by the 500°C heating, and at this temperature, the residual intensity was at least 80% of the NRM intensity (this does not mean that 80% of the NRM was thermally stable because the NRM is the resultant of two non-colinear vectors). The magnetization directions of the pilot specimens that lost more than 20% of the NRM intensity by 500°C changed with every heating and contained no unique thermally resistant direction. On the basis of this observation, all remaining Monument Valley specimens were heated once to 500°C and specimens with less than 80% of the NRM intensity were not included in the paleomagnetic pole calculations. The directions of the samples that kept more than 80% of the NRM intensity were assumed to be unbiased samples of the primary magnetization and hence suitable for the computation of a paleomagnetic pole. The use of this criterion eliminated the necessity of following in detail the changes with temperature in the direction of every specimen.

The validity of this procedure for identifying the Halgaito specimens with a stable primary

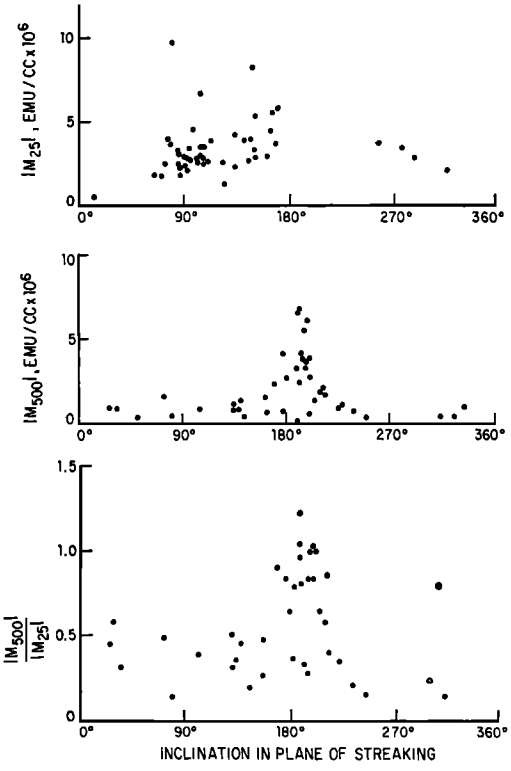


Fig. 6. Magnetization intensity of Halgaito tongue specimens plotted against inclination in the plane of streaking. Upper plot, NRM intensities; middle plot, intensities after heating to 500°C; lower plot, relative intensities after the heating.

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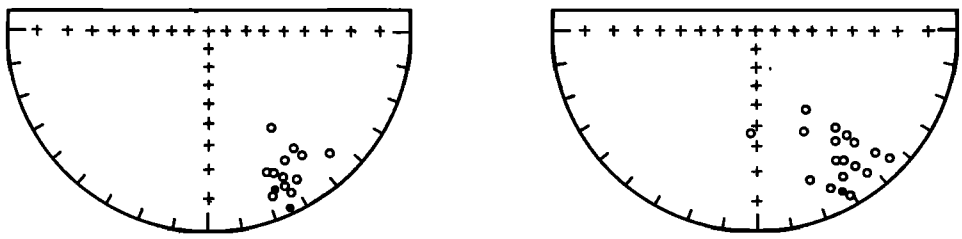


Fig. 7. Magnetization directions of the stable Monument Valley specimens after heating to 500°C.

magnetization is illustrated in Figure 6, plots of the NRM intensity, 500°C intensity, and the intensity ratio versus inclination in the plane of streaking. The specimens with a large relative intensity are narrowly distributed, whereas the weaker specimens are broadly scattered. Some specimens are not shown on the lower two plots because the magnetization became too weak to be reliably measured.

Of the 308 demagnetized Monument Valley specimens only 32 from the Hoskinnini and Halgaito satisfied the stability criterion. Figure 7 shows the direction of the residual magnetization of the specimens with a thermally stable component of the NRM, and in Table 2 the mean directions, precision parameters, and pole positions are indicated. No samples from the De Chelly sandstone, Organ Rock tongue, Cedar Mesa sandstone, and Rico transition contained a thermally stable Permian magnetization.

What about the majority (90%) of the Monument Valley specimens that were discarded in the final analysis? The NRM of most of these specimens is explicable as the vector sum of a recent component and a Permian component, although some Halgaito directions (Figure 3*b*) diverged greatly from the plane of streaking and the Organ Rock tongue (Figure 3*d*) is clustered slightly southwest of the expected position. On demagnetization, however, the directions did not remain in the plane of streaking with inclinations between the axial dipole and Permian inclinations, as one would expect, but they scattered erratically over both hemispheres. Figure 4 shows the path of one specimen and the effect is also apparent in Figure 6, where many discarded specimens ($|M_{600}|/|M_{25}| < 0.8$) have inclinations outside the predicted range (60°–190°). We cannot explain the behavior of these specimens and do not know the significance of the weak high-temperature magnetization. (A reviewer suggested the scattered directions could be explained by the theory of *Roy and Robertson* [1968]. Their model requires that each specimen contain components of magnetization with both polarities. We found no direct evidence for a Permian field of normal

TABLE 2. Results

Mean directions and precision parameters computed with Fisher's techniques [*Irving*, 1964]. Specimens given unit weight.

Formation	No. of Stable Specimens	Declination, deg.	Inclination, deg.	k	α_{95}	Pole Position
Hoskinnini (Triassic?)	18	150	−16	23	7	50°N, 121°E
Toroweap	11	157	+3	36	7	47°N, 103°E
Hermit	4	161	−7	76	8	53°N, 101°E
Halgaito	14	151	−11	54	6	49°N, 117°E

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polarity but cannot disprove their model. *McMahon and Strangway* [1968] summarize the upper Paleozoic polarity data.)

Grand Canyon Results

Most of the Grand Canyon and all the northwestern Arizona specimens were too weakly magnetized to permit demagnetization experiments. Exceptions were specimens from the Toroweap formation and Hermit shale collected on the Hermit Trail; all these specimens (15) were progressively demagnetized. In every case the magnetization vector moved in the plane of streaking toward the southeast, and over the first few hundred degrees the component of magnetization that was destroyed was parallel to the axial dipole direction (as in Figure 5). The intensity of magnetization decreased steeply during the first few heatings but remained constant above 400°C with a magnitude of 50 to 80% of the NRM intensity. The direction also remained constant over the last few heatings, evidence that the unstable component had been completely removed. Figure 8 shows the NRM for these specimens and the directions of magnetization remaining after heating to 630°C. Table 2 gives the final results for the thermally stable component of magnetization and the corresponding geomagnetic pole positions. Although only four specimens from four samples of the Hermit shale were measured, the results are included because a discrete component could be isolated, and it is not significantly different from the Toroweap results from the same locality.

Ten of the most strongly magnetized Supai specimens were heated once to 400°C and remeasured. In all cases the magnetization vectors (see Figure 9) moved toward the southeast and the intensities decreased by more than 70%. Thus, the Supai formation also has acquired a large secondary magnetization and the measurements of the NRM do not indicate the ancient field direction. Even specimens from the Hermit shale outcrop along the Kaibab Trail (12 km east of the Hermit Trail sampling site) could not be successfully demagnetized because they contained no thermally stable direction.

Secondary Magnetization

There are two principal mechanisms by which these rocks could have acquired a secondary magnetization along the present field direction. The thermally unstable component could be either a chemical magnetization created by a recent in situ growth of magnetic particles above

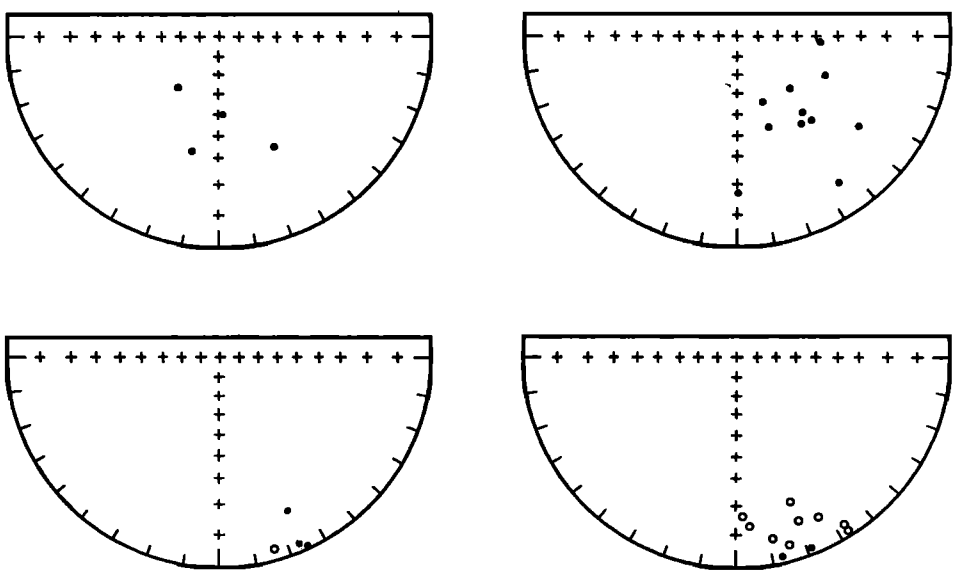


Fig. 8. Magnetization directions of Grand Canyon formations. Upper figures, NRM; lower figures, directions after heating to 630°C.

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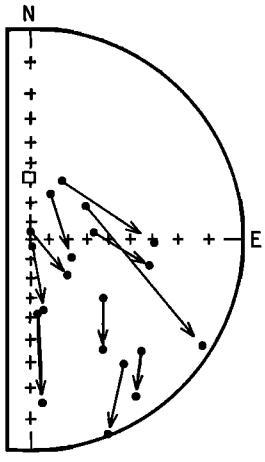


Fig. 9. Magnetization of ten Supai specimens before and after heating to 400°C.

the critical grain size, or a viscous magnetization due to the spontaneous acquisition of a magnetic moment through the thermal fluctuations in the moments of individual grains (*Nagata* [1961] gives a complete discussion of CRM on pages 196–221 and of VRM on pages 245–259). Undoubtedly, both mechanisms have affected the sediments. A viscous magnetization is often present in paleomagnetic samples, although in many formations its magnitude is negligible, and weathering of exposed outcrops could easily produce a chemical magnetization through the solution and reprecipitation of the hydrous iron oxides. We can make some conclusions about the secondary magnetization from blocking temperatures observed on the thermal decay curves (plots of intensity versus temperature).

The blocking temperature of a single magnetic grain is the temperature at which the magnetic relaxation time (proportional to exp $(\alpha V/T)$, where V is the grain volume, T is temperature, and α is a function of the magnetic properties of the particle [*Néel*, 1955]) is of the order of the duration of the experiment, say 10 minutes. The blocking temperature thus determines the time constant of the magnetization of any lower temperature, assuming that the other parameters are constant. This relationship has been conveniently plotted by *Chamalaun* [1964]. If there are a large number of grains in a rock sample with similar blocking temperatures, the thermal decay curves will show an abrupt change in intensity over a narrow range around an average specimen blocking temperature. Of course, during a thermal demagnetization, the moments of some grains are randomized by each heating, for there is usually a continuous distribution in grain size (V). At the specimen blocking temperature, however, a relatively large proportion of the NRM is destroyed.

The thermal decay curves for the Toroweap and Hermit showed a large decrease in intensity around 150°C. This is a characteristic blocking temperature since measurements of the induced magnetization (2000-gauss field) as a function of temperature were nearly reversible and indicated only a hematite Curie point above 650°C. A time constant of 10 minutes at 150°C is equivalent to less than a year at 20°C. Thus, this component, found to be randomly directed, was a viscous magnetization acquired since the sampling.

Some Toroweap and Hermit specimens exhibited a second blocking point near 450°C. This feature did not appear on the Monument Valley thermal decay curves, but we have already seen (Figures 4 and 5) that part of the unstable magnetization for some of these specimens resisted the 400° heating, but not the 500°C. Although a true blocking point is absent, certainly a significant fraction of the unstable magnetization had a relaxation time greater than 10 minutes at ~450°C. Furthermore, this component was probably directed along the present field direction.

It seems unlikely that this part of the secondary magnetization is a viscous component, if the simple theory is applicable. If it is assumed that this part was viscous and grew since the most recent field reversal, the time of exposure is less than 10^6 years. With this value as the relaxation time, an ambient temperature of 120°C is required. There is no climatological evidence for such a rise in surface temperature, and depths of a few kilometers are needed to reach this temperature inside the earth. It is possible that this thermally resistent secondary magnetization was acquired in a remote normal polarity epoch that lasted much longer than 10^6 years, but this interpretation seems unlikely. A more reasonable alternative is that the part of the secondary magnetization that resisted the 400°C heating is a chemical rather

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than viscous magnetization and is less than a million years old.

It is perhaps surprising that such a small proportion (~10%) of the samples could be successfully demagnetized, either because the Permian component was completely absent, or because its thermal stability was so low it could not be separated from the recent magnetization. Possibly the same weathering phenomenon that created the unstable magnetization simultaneously destroyed or altered the mineral grains that were magnetized along the Permian field direction.

Comparison with Other Permian Results from North America

One of the puzzling features of previous Permian results from the Southwest is the large scatter in the geomagnetic poles. From the demonstrated instability in the NRM of all the rocks measured in this study, it is likely that most of the sediments in the area have acquired a recent secondary magnetization. Indeed, the latitudes of the poles given here are higher than any previous results from the Colorado Plateau, the expected effect if the NRM is the vector sum of a Permian magnetization and a variable recent magnetization. The scatter in the previous poles was, however, in longitude as well as latitude. This cannot be due to the secondary magnetization for the recent secular variation amplitude is too small. The longitudinal scatter is perhaps present in our results, the Grand Canyon pole lying 15° east of the Monument

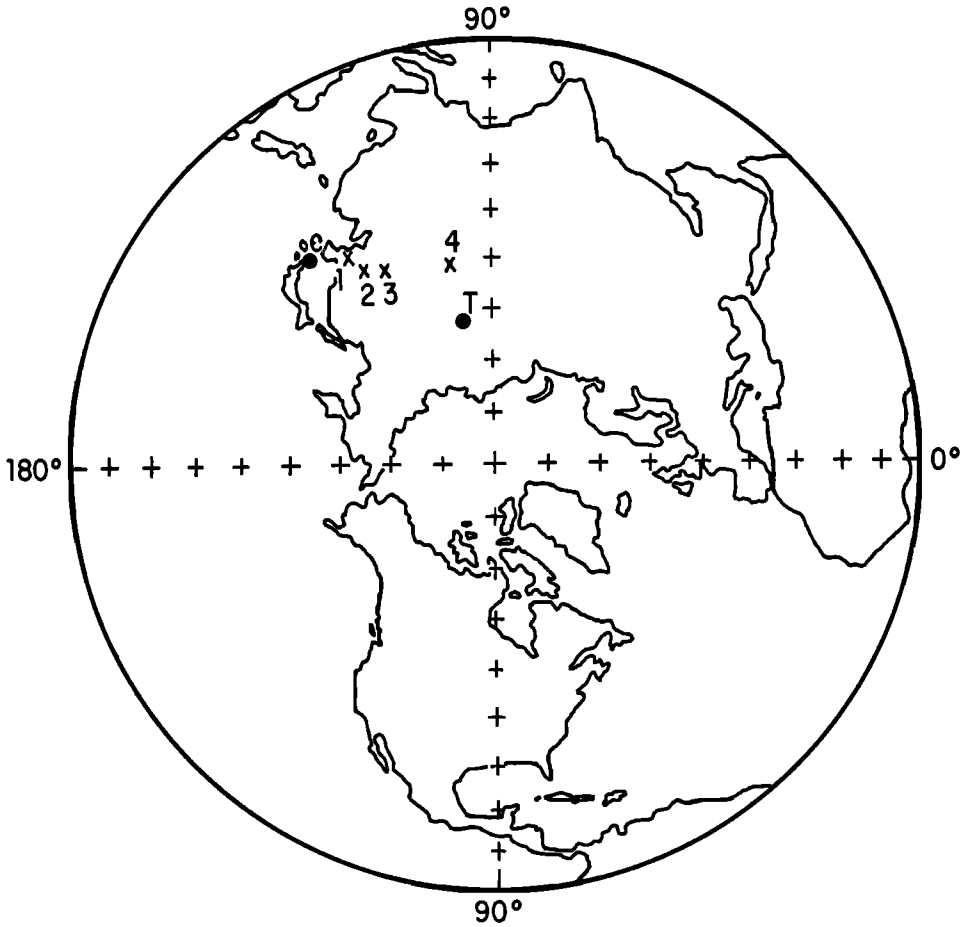


Fig. 10. North American paleomagnetic pole positions for the Carboniferous (C), Permian (X), and Triassic (T). (1) Eastern Canada [*Black,* 1964; *Roy,* 1963, 1966]; (2) West Virginia [*Helsley,* 1965]; (3) Monument Valley; (4) Grand Canyon.

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Valley one, but whether the divergence is a real feature of the Permian field or a variability due to an insufficient number of sites can only be determined by further sampling and demagnetization.

Finally, Figure 10 compares the geomagnetic poles for the Grand Canyon and Monument Valley sediments with other paleomagnetic results from North America. The mean Carboniferous (C) pole is similar to that computed by *Irving* [1964] but includes recent results from eastern Canada by *Black* [1964]; the Triassic pole (T) is taken from *Beck* [1965]. Because of the instability found in all the rocks measured in this work, it is concluded that the previous results from the Permian of the Southwest, including the results from the Supai formation, are not reliable indicators of the Permian field direction. The results presented here agree well with the results of *Helsley* [1965], who used ac demagnetization on the Dunkard series in West Virginia, and with the eastern Canada data of *Roy* [1963, 1966], who applied thermal demagnetization to his specimens.

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